Lending Game – Stage 1: the final pricing rule

Pricing the pool we actually win – Team "Lender 2"

Internal note; supersedes the band section of the stage-1 strategy note

14 October 2026

1 The rule

For applicant i we estimate the probability of default $\hat{p}_i = \mathbb{E}[D_i \mid X_i, s_{2,i}]$ with gradient-boosted trees (employment, marital status, income, log income, our signal s_2, a "no signal" indicator; out-of-sample AUC 0.735, calibrated within 0.4 points per decile). Income defines six segments s with a margin $m(s)$ and a threshold $\tau(s)$ (Table 1). The offered rate is

$$r_i \;=\; \begin{cases} \dfrac{\hat{p}_i}{1-\hat{p}_i} + m(s_i) & \text{if } \hat{p}_i \le \tau(s_i) \quad \text{(the safe part)} \\[2ex] \min\Big(\dfrac{\tilde{\pi}(\hat{p}_i)}{1-\tilde{\pi}(\hat{p}_i)} + X,\; 0.989\Big) & \text{if } \hat{p}_i > \tau(s_i) \quad \text{(the band)} \end{cases} \qquad X = 0.05, \tag{1}$$

where $\tilde{\pi}(\hat{p})$ is the *selection-adjusted default rate*: the default rate of the applicants we actually win when every lender prices on its own model (Section 2). The two pieces are joined over one point of PD at the threshold, $r = r_{\text{safe}} + w(\hat{p})\,(r_{\text{band}} - r_{\text{safe}})$ with $w(\hat{p}) = \sigma\big(6(\hat{p}-\tau)/0.01 - 3\big)$, so that no applicant faces a cliff. Rates are rounded to four decimals; every applicant receives an offer.

Income segment	0	1–5k	5–10k	10–20k	20–40k	40k+
margin $m(s)$ on the safe part	4.0%	3.5%	3.5%	3.0%	3.0%	3.0%
threshold $\tau(s)$	12%	12%	10%	8%	25%	25%
share of the segment in the band (new file)	92%	74%	69%	18%	0%	0%

Table 1: Segment parameters (unchanged from the first submission; chosen by maximizing expected profit over competitor scenarios on the past loans). Low thresholds at low income because that is where the rivals' hidden information is large (standard deviation of $p_{\text{full}} - \hat{p}$: 7 points at zero income, 0.3 points above 40k).

In words: on applicants whose risk our model can price competitively we charge break-even plus a thin margin and aim at our own third of the market; on the rest, where the other lenders' private signals create a winner's curse, we charge the break-even *of the people we would win*, plus five points, never above 98.9%.

2 Where $\tilde{\pi}$ comes from: the break-even given that we win

The problem. $\hat{p}$ is calibrated on the population, but we only lend to the applicants who *accept*. An applicant takes the lowest offer; the rivals see signals we do not and price higher exactly when those signals are bad. So the applicants who accept our offer are, on average, riskier than $\hat{p}$ – the winner's curse – and the gap grows with risk because the rivals' hidden information does. Pricing at $\hat{p}/(1-\hat{p})$ + a few points therefore loses money on every category above ∼10% PD (realized default of the loans we win: 15% at $\hat{p} = 12\%$, 24% at 17%, 32% at 23%, 42% at 27%).

The measurement. The past-loans file contains all three signals, so we can reproduce the selection exactly. For every past loan i we compute the three lenders' out-of-sample estimates $\hat{p}_{k,i} = \mathbb{E}[D \mid X, s_k]$, $k = 1, 2, 3$, and the full-information probability $p_i^{\text{full}} = \mathbb{E}[D \mid X, s_1, s_2, s_3]$. We then let every lender quote $\hat{p}_{k,i}/(1 - \hat{p}_{k,i}) + m(s_i)$ – the same rule, each on its own information – and apply the choice rule of the game for each applicant type $t \in \{1, 2, 3\}$ (one third each, the preferred lender's rate reduced by 0.02):

$$\omega_i \;=\; \frac{1}{3}\sum_{t=1}^{3} \mathbf{1}\Big\{\, r_{2,i} - 0.02 \cdot \mathbf{1}[t=2] \;<\; \min_{k\neq 2}\big(r_{k,i} - 0.02 \cdot \mathbf{1}[t=k]\big)\Big\} \tag{2}$$

is the probability that applicant i takes our offer. For each PD bin B (one point wide),

$$\tilde{\pi}(B) \;=\; \frac{\sum_{i\in B}\omega_i\, p_i^{\text{full}}}{\sum_{i\in B}\omega_i}, \qquad \tilde{\pi}^{\text{real}}(B) = \frac{\sum_{i\in B}\omega_i D_i}{\sum_{i\in B}\omega_i}, \qquad \tilde{\pi}^{95}(B) = \tilde{\pi}^{\text{real}} + 1.645\sqrt{\frac{\tilde{\pi}^{\text{real}}(1-\tilde{\pi}^{\text{real}})}{\sum_{i\in B}\omega_i}}. \tag{3}$$

$\tilde{\pi}$ is the default rate of the applicants we win, and $\tilde{\pi}/(1 - \tilde{\pi})$ is the rate at which winning them still breaks even. The expected version (with p^{full}) is smooth and is the one used in the rule; the realized version and its 95% upper bound check it (Table 2, Figure 1A).

Our PD $\hat{p}$	Share we win	$\tilde{\pi}$ (expected)	$\tilde{\pi}^{95}$ (realized, upper)	Penalty $\tilde{\pi} - \hat{p}$	Plain break-even	Adjusted break-even
5%	44%	5.1%	5.5%	+0.6 pt	4.7%	5.3%
8.5%	39%	10.2%	11.3%	+1.7	9.2%	11.3%
10.5%	38%	12.7%	14.2%	+2.1	11.7%	14.5%
12.5%	44%	14.4%	15.8%	+1.9	14.3%	16.9%
14.5%	29%	18.2%	19.9%	+3.7	16.9%	22.2%
16.5%	17%	22.9%	26.7%	+6.4	19.7%	29.6%
18.5%	18%	24.1%	27.9%	+5.6	22.7%	31.7%
20.5%	11%	26.9%	30.1%	+6.5	25.7%	36.8%
22.5%	18%	29.0%	37.0%	+6.5	29.1%	40.8%
24.5%	13%	35.5%	39.7%	+11.0	32.5%	55.0%
26.5%	4%	46.8%	58.2%	+20.2	36.2%	88.1%
$\geq 28\%$	$\leq 1\%$	$> 50\%$	–	> 20	$\geq 40\%$	none below the cap

Table 2: The winner's curse per PD level, measured on the past loans when every lender prices on its own model.

Three remarks. (i) The selection – and hence $\tilde{\pi}$ – is the same under *any* common monotone transformation of the rule: if every lender applies the same increasing function to its own PD, the same applicants are won. So the curve does not depend on the margin the rivals use, only on their models being as informative as ours. (ii) The curse is small where our model can compete (below 10% PD the penalty is at most 2 points, which the 3–4% margin covers) and explodes above 25%: beyond ∼27% the applicants we would win in competition default more than half the time, no finite rate breaks even, and the rule reaches the cap – those offers (2% of applicants) are a residual lender-of-last-resort offer, taken only if both rivals refuse, not a bet. (iii) The 95% upper bound stays within 1–4 points of the expected curve up to 25% PD, so the five-point margin also covers the estimation uncertainty there.

3 Why a margin X, and how it was chosen

At $X = 0$ we would break even on the band even if we won all of it. The margin is what we earn on the band, but every point of margin is also a point by which a rival pricing the same way can undercut us: with the 2% preference, whoever is lowest within two points takes the whole band. So X is a bet on the rivals' margins, and we chose it by Monte Carlo.

The simulation. Each rival is drawn at random from a profile of plausible behaviours and priced on its own PD estimate; the market is evaluated exactly (the three applicant types with weight 1/3

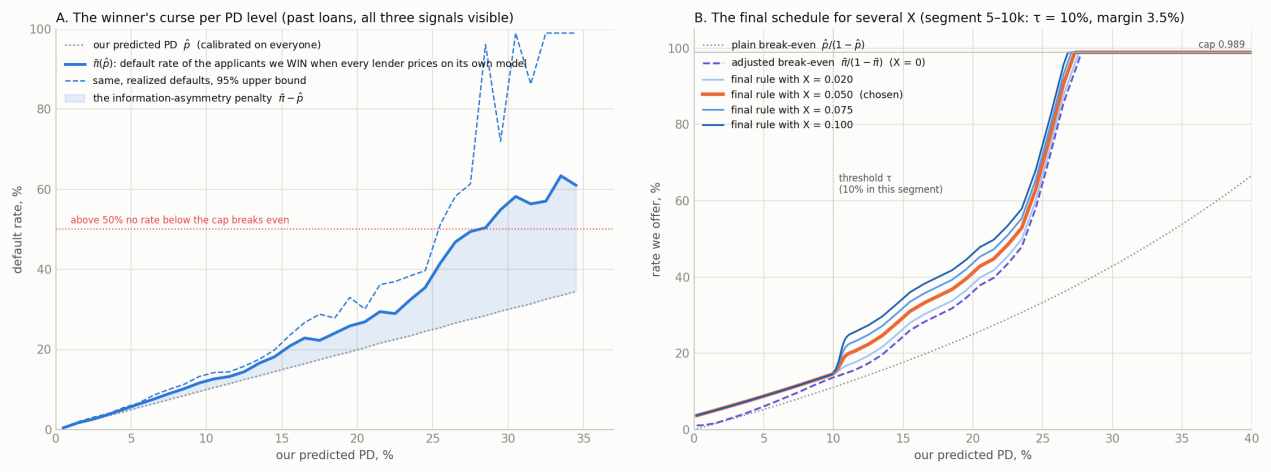


Figure 1: A: our predicted PD versus the default rate of the applicants we win (expected, and realized 95% upper bound); the shaded area is the price of the information asymmetry. B: the final schedule in a typical low-income segment (threshold 10%, margin 3.5%) for several margins X; $X = 0.05$ is the submitted rule.

each, ties split evenly among tied lenders – an adverse-tie variant gives identical results, since the schedules create no exact ties), rates rounded as in the file, profits computed with p^{full}. Profiles and their weights (our belief that the rivals are curse-aware and do not use a lender-of-last-resort rate): *like us, no net* – rivals pricing the band at their own adjusted break-even plus a margin drawn between 3 and 25 points (40%); *smart* – curse-aware rivals using a net, a ramp, a large margin, a data-driven schedule, refusal or a compromise (20%); *clones* of our method with other parameters (10%); *best-responders* who undercut our data-driven schedule (10%); a broad *mixed* profile including naive behaviours (10%); pure *netters* (5%); *model-followers* pricing model + 2–6% for everyone (5%). 120 independent rival pairs per profile.

Band schedule	Prior-weighted over the seven profiles mean profit	P(first)	P(last)	"Like us, no net" rivals only mean profit	P(first)	P(last)
adjusted break-even, $X = 0$	11.9 M€	79%	2%	9.7	62%	4%
$X = 0.02$	13.6	82%	1%	11.0	70%	2%
$X = 0.05$ **(submitted)**	**15.7**	**75%**	**3%**	**12.4**	**55%**	**6%**
$X = 0.075$	17.1	70%	5%	12.9	47%	11%
$X = 0.10$	18.3	65%	7%	13.1	36%	17%
$X = 0.15$	19.6	54%	15%	12.2	14%	35%
ramp 0.45 → 0.65 (previous candidate)	22.2	42%	41%	8.2	2%	91%
flat net 0.828 (first file)	11.0	13%	68%	6.7	0%	98%

Table 3: Expected profit and rank among the three lenders as a function of the margin X (120 random rival pairs per profile). Profit rises with X; the probability of finishing first falls with it.

The criterion. Stage-1 performance is graded on the level of profit *and* on the ranking, so we declared the rule: maximize the prior-weighted expected profit subject to a probability of finishing last of at most 3%. It selects $X = 0.05$: 15.7 M€ on average, first three times out of four, last 3% of the time, never below 7.2 M€ on average in any profile. Higher margins buy profit at the cost of rank (at $X = 0.10$, 18.3 M€ but last 7% – 17% against rivals who price like us); the hand-made ramp earns more on average only because of jackpots against netters, and finishes last in 41% of the draws (91% against rivals who price like us). Figure 2 shows the dial.

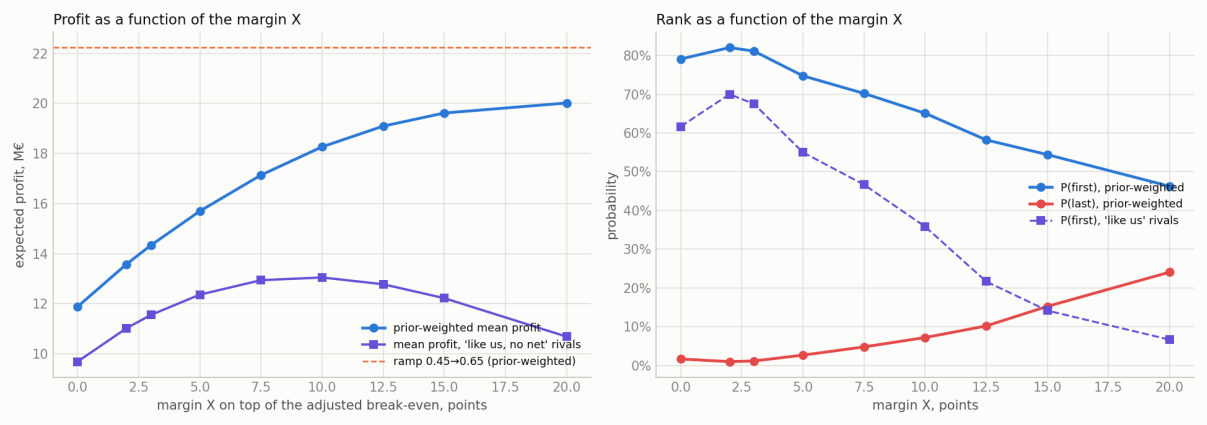


Figure 2: Expected profit (left) and rank (right) as a function of the margin X on top of the adjusted break-even.

4 The submitted file

`output/round1/rates_round1.csv`, built from the model refitted on all past loans (full-precision PDs) by

```
python make_submission.py --adjusted-x 0.05 --cap 0.989 --phase-in 0.01
```

100,000 rows, columns `id` and `rate`, ids 1 to 100,000 in order and unique, no missing value, all rates in $[0.0304, 0.989]$; the SHA-256 checksum (`c9c7f714...`) and the exact parameters are stored in `rates_round1_params.json`.

rate	≤5%	5–10%	10–15%	15–20%	20–30%	30–40%	40–50%	50–60%	60–80%	80–98.9%
offers	13,479	26,393	13,208	9,019	15,242	10,877	3,576	3,559	1,515	3,132

Table 4: Composition of the file: 54,418 safe-part offers between 3.0% and 17.6%; 45,582 band offers between 11.9% and 98.9%, of which 2,027 at the cap.

5 What we expect, and what could go wrong

Across the rival profiles the expected profit of this rule is 7–16 M€ (median round 6–8 M€), finishing first in about three quarters of the simulated markets. The two assumptions that matter most are (i) that the rivals' models are about as informative as ours – if theirs are worse, the curse is smaller and we are slightly over-pricing the band; if better, the five-point margin is our cushion – and (ii) the rules as written: an applicant with at least one offer always borrows, which is what makes the capped offers harmless. After stage 1 we will read the rivals' actual band prices from the results and re-price each PD category at their observed level minus a few points where that stays above the adjusted break-even, which is the only part of the rule that depends on a belief rather than on a measurement.